Summary and Recommendations

1. Objective

The primary objective of this project is to analyze the **Telco Customer Churn dataset** using Exploratory Data Analysis (EDA) techniques to identify the key factors influencing customer churn. The project aims to uncover patterns in customer demographics, service subscriptions, contract types, payment methods, tenure, and monthly charges to generate actionable business insights that can help improve customer retention, reduce churn, and support data-driven decision-making.

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2. Key Insights

- The dataset shows an overall **customer churn rate of 26.54%**, while **73.46%** of customers remain with the company.
- Customers with **Month-to-Month contracts** contribute to the highest percentage of churn compared to customers with long-term contracts.
- **Senior citizens** have a relatively higher churn rate than non-senior customers.
- Customers with **shorter tenure** are more likely to leave the company, whereas long-term customers exhibit greater loyalty.
- **Fiber Optic Internet users** experience higher churn than DSL users and customers without internet services.
- Customers who do **not subscribe to Online Security, Device Protection, Tech Support, or Online Backup** services are significantly more likely to churn.
- Higher monthly charges are generally associated with increased churn.
- Services such as **Streaming TV, Streaming Movies, Multiple Lines, and Phone Service** have comparatively less impact on customer churn.

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3. Churn Analysis

- **Total Churn Rate: 26.54%**
- **Customer Retention Rate: 73.46%**
- Approximately **1 out of every 4 customers** has discontinued the company's services.
- Month-to-Month contract customers represent the highest-risk customer segment.
- Customers with low tenure and higher monthly charges show the greatest probability of churning.
- Long-term contract customers demonstrate significantly better retention.

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4. Tenure Analysis

- Customer tenure is one of the strongest indicators of customer loyalty.
- Customers with **lower tenure (new customers)** have the highest churn probability.
- Customers who remain with the company for several years are much less likely to churn.
- The analysis indicates that the **first few months of the customer lifecycle are the most critical** for customer retention.
- Improving customer experience during the onboarding phase can substantially reduce future churn.

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5. Payment Method Analysis

- Customers using **Electronic Check** have the highest churn rate among all payment methods.
- Customers using **Automatic Bank Transfer** and **Credit Card (Automatic Payment)** tend to have much lower churn.
- Automatic payment methods indicate higher customer engagement and commitment.
- Encouraging customers to switch from Electronic Check to automatic payment options may improve retention.

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6. Recommendations

Based on the findings of the analysis, the following recommendations are proposed:

- **Target High-Risk Customers:** Focus retention campaigns on customers most likely to churn, particularly the **26.54% churn-risk segment**.
- **Promote Long-Term Contracts:** Encourage customers to migrate from Month-to-Month plans to One-Year or Two-Year contracts through discounts and loyalty benefits.
- **Improve Customer Onboarding:** Provide personalized support during the first few months of service to reduce early churn.
- **Enhance Fiber Optic Services:** Investigate customer complaints related to Fiber Optic services and improve service quality and pricing where necessary.
- **Promote Value-Added Services:** Encourage customers to subscribe to Online Security, Device Protection, Tech Support, and Online Backup services to increase customer satisfaction.
- **Encourage Automatic Payments:** Offer incentives for customers to adopt automatic payment methods instead of Electronic Checks.
- **Focus on Senior Citizens:** Design personalized customer engagement programs for senior citizens to improve retention.
- **Monitor High Monthly Charges:** Review pricing strategies for customers with higher monthly bills and provide customized offers where appropriate.

- **Leverage Predictive Analytics:** Develop machine learning models to identify customers at risk of churning and implement proactive retention strategies before they leave.